

with the message "This trunk is empty," and Trunk 3 is inscribed with the message "The treasure is in Trunk 2." The Queen, who never lies, tells you that only one of these inscriptions is true, while the other two are wrong. Which trunk should you select to win?

Soln: $P_1 \rightarrow$ Trunk 1 tells the truth $P_2 \rightarrow$ Trunk 2 tells the truth
 $P_3 \rightarrow$ Trunk 3 tells the truth

Queen: $(P_1 \wedge \neg P_2 \wedge \neg P_3) \vee (\neg P_1 \wedge P_2 \wedge \neg P_3) \vee (\neg P_1 \wedge \neg P_2 \wedge P_3)$ is True

never lies. Exactly one inscription is True.

Suppose P_1 is True. By defn. of negation, $\neg P_1$ is False. By defn. of AND, $(\neg P_1 \wedge P_2 \wedge \neg P_3)$ and $(\neg P_1 \wedge \neg P_2 \wedge P_3)$ are respectively False. By defn. of OR, $(P_1 \wedge \neg P_2 \wedge \neg P_3)$ is True. By defn of AND, $\neg P_2, \neg P_3$ are respectively True. By defn of negation P_2 and P_3 are False.

P_4 : This trunk is empty (1) P_5 : This trunk is empty (2)

P_6 : The treasure is in trunk 2.

~~P_1 must be True~~ Since P_1 is True, Trunk 1 is empty.
 P_2 is False. $\therefore$ The treasure is in Trunk 2, since

Trunk 2 is not empty. That means P_3 must be True. This is a contradiction to the fact that P_3 is False. $\therefore P_1$ is False. So, Trunk 1 is not empty and it contains the treasure.

Let P_2 be True. $\therefore$ Trunk 2 is empty (consistent)

$\therefore P_3$ must be False. $\therefore$ The treasure is in Trunk 2 is False (consistent)

Let P_2 be False. $\therefore$ Trunk 2 contains the treasure. This is a contradiction to the fact that Trunk 1 contains the treasure.

$\therefore P_1$ is False, P_2 is True, P_3 is False. Trunk 1 contains the treasure